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SUPPLEMENTARY MATERIAL

The Spatial Geometry of the Lorentz Tetrad

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Abstract
We realise the four legs of a Lorentz tetrad as vectors in the three-dimensional spatial sector and ask which configurations are possible. Four premises, each a transcription of the tetrad, suffice: every pair of legs bears the same relation, as $\eta_{ab} = 0$ for every pair; every leg has the same magnitude, as $|\eta_{aa}| = 1$; the legs are distinct; and the realisation lives in $\mathbb{R}^3$. They force the Gram matrix $G = c^2 I - (c^2/3)(J - I)$, and hence the regular tetrahedron, unique up to $O(3)$. Closure $\sum_a v_a = 0$ is derived, not assumed. Three is the only dimension in which such a realisation both exists and is forced. The realisation keeps the pattern of η's entries, not their values, which no four vectors in $\mathbb{R}^3$ reproduce; it sets aside the sign of η_{00}, which no squared length can carry, and this is the content of the premise of equal magnitude. Without that premise an observer's projection, in which the time leg vanishes, is admitted. Each premise is shown to be necessary. Spatial parity gives the second tetrahedron, the cube and the stellated octahedron.

Keywords: tetrad, Gram matrix, regular tetrahedron, stellated octahedron, proof

Setting

A Lorentz tetrad has four legs e_a, $a = 0, 1, 2, 3$, linearly independent, with $\eta(e_a, e_b) = \eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$. A *spatial realisation* is a 4-tuple $V = (v_0, v_1, v_2, v_3)$ of vectors $v_a \in \mathbb{R}^D$, one per leg, with Gram matrix $G_{ab} = v_a \cdot v_b$. We write I for the 4×4 identity, J for the all-ones matrix and $\mathbf{1} = (1, 1, 1, 1)^T$.

Premises

(R) Equal relation. $v_a \cdot v_b$ takes one value for all $a \neq b$. In the tetrad $\eta_{ab} = 0$ for every pair, time with space and space with space alike. The common value 0 cannot be kept, since four mutually orthogonal non-zero vectors do not fit in $\mathbb{R}^3$; its being common to all pairs can.

(M) Equal magnitude. $|v_a| = c > 0$ for all a. In the tetrad $|\eta_{aa}| = 1$ for every leg. A diagonal entry of a Gram matrix is a squared length and cannot be negative, so no realisation reproduces $\eta_{00} = -1$; (M) keeps its magnitude, which is that of every other leg.

(N) Distinct legs. The v_a are not all equal, as the legs are linearly independent.

(D) Spatial sector. $D = 3$, the number of positive entries of η.

Together, (M) and (R) say that G is invariant under every permutation of the legs, since the orbits of S_4 on ordered pairs are the diagonal and the off-diagonal pairs. Closure is not assumed.

Rigidity

Lemma. *If $U, W \in \mathbb{R}^{D\times 4}$ have $U^T U = W^T W$ of rank D, then $W = QU$ for a unique $Q \in O(D)$.*

Proof. For every $\lambda \in \mathbb{R}^4$, $|U\lambda|^2 = \lambda^T U^T U \lambda = |W\lambda|^2$. Hence $U\lambda = U\lambda'$ implies $W\lambda = W\lambda'$, and $Q : U\lambda \mapsto W\lambda$ is a well-defined linear isometry from the column space of U onto that of W. Both are $\mathbb{R}^D$ because the rank is D, so $Q \in O(D)$, and it is fixed by its values on the columns. □

The Forced Gram Matrix

Theorem. *Under (R), (M), (N) and (D), the legs close, $\sum_a v_a = 0$, and*

$$\boxed{G = c^2 I - \frac{c^2}{3}(J - I),} \tag{1}$$

that is $G_{aa} = c^2$ and $G_{ab} = -c^2/3$ for $a \neq b$.

Proof. By (M) and (R), $G = c^2 I + \beta(J - I) = \alpha I + \beta J$ with $\alpha = c^2 - \beta$. Since $J\mathbf{1} = 4\mathbf{1}$ and J vanishes on $\mathbf{1}^\perp$, G has eigenvalue $\alpha + 4\beta$ on $\mathbf{1}$ and α, three times, on $\mathbf{1}^\perp$. Every pair of legs is at the same distance, $|v_a - v_b|^2 = G_{aa} + G_{bb} - 2G_{ab} = 2\alpha$, so by (N) $\alpha > 0$ and $\mathrm{rank}\, G \geq 3$. By (D), $\mathrm{rank}\, G \leq 3$. Hence $\alpha + 4\beta = 0$, so $G\mathbf{1} = 0$ and $|\sum_a v_a|^2 = \mathbf{1}^T G \mathbf{1} = 0$: the legs close. Then $c^2 = \alpha + \beta = 3\alpha/4$ and $\beta = -\alpha/4 = -c^2/3$, which is Eq. (1). □

Corollary 1 (the regular tetrahedron). *The four points v_a are pairwise equidistant, with edge $c\sqrt{8/3}$ and centroid at the origin: they are the vertices of a regular tetrahedron of circumradius c, and the angle between any two*

legs is $\arccos(-1/3)$. *With* $c = \sqrt{3}$ *the columns of*

$$M = \begin{pmatrix} +1 & +1 & -1 & -1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix} \tag{2}$$

realise Eq. (1)*, and every realisation is* $V = (c/\sqrt{3})\,QM$ *for a unique* $Q \in O(3)$.

Proof. The distances are $2\alpha = 8c^2/3$ and closure puts the centroid at the origin. Each column of M has three entries ± 1, and each pair of columns agrees in exactly one place, so $M^T M = 3I - (J - I)$. Both V and $(c/\sqrt{3})M$ have Gram matrix (1), of rank 3, and the Lemma gives Q. □

Corollary 2 (why three dimensions). *For* K *legs satisfying (R), (M) and (N) in* $\mathbb{R}^D$*, a realisation exists if and only if* $D \geq K - 1$*, and it is forced to the closed regular simplex if and only if* $D = K - 1$*. For the tetrad,* $K = 4$ *and* $D = 3$.

Proof. The argument above with K in place of 4 gives eigenvalues $\alpha > 0$ ($K - 1$ times) and $\alpha + K\beta$, so $\operatorname{rank} G \geq K - 1$ and no realisation exists for $D < K - 1$. The regular simplex lies in $\mathbb{R}^{K-1} \subseteq \mathbb{R}^D$, so one exists for $D \geq K - 1$. If $D = K - 1$, then $\alpha + K\beta = 0$ and the simplex is forced. If $D \geq K$, any $\alpha + K\beta > 0$ is also realisable, for example $\beta = 0$: K orthogonal legs of equal length. □

Corollary 3 (parity completion). *Spatial parity acts on the spatial sector as* $-I$*, so it sends every realised vector to* $-v_a$ *and preserves* G*:* $-V$ *is a second realisation.* V *and* $-V$ *share no point, and for* $V = M$ *their eight points are the vertices of the cube* $\{-1, +1\}^3$*; the compound of the two tetrahedra is Kepler's stella octangula [1, 2], the stellated octahedron.*

Proof. $(-V)^T(-V) = V^T V$. A shared point $v_a = -v_b$ would need $G_{ab} = -c^2$, but $G_{ab} \in \{c^2, -c^2/3\}$. The columns of M and $-M$ are eight distinct sign triples, which exhaust $\{-1, +1\}^3$. □

Necessity of the Premises

Removing any one premise admits configurations that satisfy the other three but are not the regular tetrahedron:

Removed	Admitted configurations
(R)	$(v, w, -v, -w)$ with $\lvert v\rvert = \lvert w\rvert$, $v \perp w$: a square, even closed
(M)	$v_0 = 0$, $v_i = c\,\hat{e}_i$: an observer's projection, in which the time leg vanishes. With every leg non-zero: $G_{aa} = (1, 3, 7, 7)\,c^2$, $G_{ab} = -c^2$, of rank 3
(N)	$v_a = v$ for all a: $G = c^2 J$
(D)	In $\mathbb{R}^4$: $v_a = c\,\hat{e}_a$, $G = c^2 I$, not closed. In $\mathbb{R}^2$: no realisation

The row for (M) shows what that premise decides. Read with the sign, η singles out the time leg, and the natural spatial picture is an observer's: three orthonormal legs and no fourth. Read by magnitude, all four legs are units, and the spatial picture is forced to the tetrahedron.

Summary

Every premise transcribes the tetrad. Equal relation is the pattern of η's off-diagonal entries, which carry no sign; equal magnitude is the pattern of its diagonal, with the sign of η_{00} set aside because no squared length can be negative; distinctness is the independence of the legs; and the dimension is the spatial sector. From these, four legs in three dimensions have exactly one configuration up to $O(3)$: the regular tetrahedron, closed, with the tetrahedral angle $\arccos(-1/3)$ between every pair. Three is the only dimension in which this configuration both exists and is forced. Parity, which the Gram matrix cannot see, supplies the second tetrahedron, and the two together form the stellated octahedron.

References

[1] J. Kepler, *Harmonices Mundi* (Linz, 1619).

[2] P. R. Cromwell, *Polyhedra* (Cambridge University Press, Cambridge, 1997).